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Research Paper

# Comparison of univariate and multivariate magnitude-squared coherences in the detection of human 40-Hz auditory steady-state evoked responses

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ABSTRACT

Objective response detection (ORD) techniques for evaluating bioelectrical evoked responses in the electroencephalogram (EEG) are based on statistical criteria rather than on visual inspection. Hence, they do not depend on human evaluation, which is often a subjective approach. Furthermore, since such techniques do not involve heuristic approaches, they may be more easily implemented and used in automatic systems. The Magnitude-Squared Coherence (MSC), together with its recently developed multivariate extension (the multiple magnitude-squared coherence – MMSC), have been pointed out as one of the most efficient ORD techniques for detecting steady-state responses in the EEG. In this work, both MSC and MMSC were applied to EEG signals collected during auditory stimulation in order to allow comparison in the detection of auditory steady-state responses (ASSRs). The stimuli consisted of 40 Hz amplitude-modulated tones delivered binaurally in the intensity of 50 dB SPL (*sound pressure level*). The best result was obtained by using MMSC in the two-electrode set C4 and Fz. This configuration led to a 0.92-detection ratio, within 111.55 s in average to detect each response and kept the false alarm ratio under 0.05. The average improvement in performance was about 11% when compared to the MSC. These results allow concluding that the detection protocol of 40 Hz ASSRs can be improved by using MMSC in multichannel EEG analysis when compared to the traditional univariate MSC approach.


## 1. Introduction

The evoked potential (EP) corresponds to an electrical manifestation of the brain response to an external stimulus [1]. They may be either transitory, whenever the response to a given stimulus vanishes completely before the next one is applied, or a steady-state one, whenever the following stimulus is applied before the response to the previous one vanishes.

The EP may be recorded on the scalp, together with the ongoing electroencephalogram (EEG). However, due to its reduced amplitude in comparison with this latter, the EP is not easily noticeable in the EEG. This occurs due to the fact that the EEG reflects other rhythmic activities from the brain, face muscles and the neck [2]. Therefore, signal processing techniques are often necessary for revealing the EP.

For the case of auditory stimulation, amplitude modulated (AM) tones lead to auditory steady-state responses (ASSRs). The most used AM tones in studies with human beings have modulating frequency within a range close to 40 Hz [3]. Such kind of stimulation results in an energy increase at the modulation frequency in the signal power spectrum [4]. Thus, ASSRs are more easily detected by means of frequency domain techniques. Assessing the presence of ASSRs in the EEG is an important task in monitoring surgeries [5], newborn auditory screening [6], as well as in brain-computer interfaces [7,8].

In this scenario, objective response detection (ORD) techniques have arisen as a way of turning the EP detection less dependent on visual inspection of averaging signals or their corresponding spectra. In such techniques, the sampling distribution of a detector is obtained under the null hypothesis of no EP in the EEG in order to establish a critical value that corresponds to a threshold for the

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detection [9]. The magnitude-squared coherence (MSC) estimate between the stimulating signal and the EEG has been suggested as an efficient ORD technique [10]. The MSC is analogous to the coefficient of determination (the square of the correlation coefficient), but in the frequency domain. Since the signal-to-noise ratio (SNR) is known to be greater in the stimulating frequency and the harmonics [11], coherence between both signals is expected to be greater in such frequencies. Furthermore, the sampling distribution of the coherence estimate is well established for Gaussian signals [12] and, for the particular case of coherence equal to zero, the statistics remain the same in spite of the sampling distribution of the second signal, provided the first one is Gaussian [13]. This leads to a very robust detector, since the false positive detection ratio will not depend on the shape of the evoked response. Furthermore, in this particular case of one periodic, deterministic signal, coherence may be estimated using only the EEG signal, which simplifies the experimental setup, since the stimulating signal need not be recorded [14].

However, for a given SNR-value, the detection ratio can only be increased at the expense of using longer stretches of EEG signals [15], which may constitute an issue in monitoring surgeries, when detection should occur as fast as possible in order to avoid injuries to the patient. With the purpose of increasing the detection ratio with coherence without augmenting the record length, the use of signals from different electrodes has been suggested to allow multiple coherence estimation [15,16]. This multivariate, spectral detector, named as multiple magnitude-squared coherence (MMSC), has been further investigated in [17] in the EEG during photic stimulation and compared to its univariate counterpart. It is worth mentioning the inexistence of published works investigating the efficiency of MMSC in the detection of ASSRs.

Thus, this work aims at evaluating the MMSC in the detection of ASSR and at establishing the more suitable electrode set that should be used. Thus, EEG signals during AM tones stimulation were collected and the detection protocol was applied based on such multivariate objective response detection (MORD) technique.

## 2. Mathematical background

### 2.1. Magnitude-squared coherence

The coherence estimated for finite-length, discrete-time signals $x[k]$ and $y[k]$, divided into $M$ windows, is given as [18]

$$\hat{\gamma}^2_{xy}(f) = \frac{\left|\sum_{i=1}^{M} X_i^*(f)\,Y_i(f)\right|^2}{\sum_{i=1}^{M} |X_i(f)|^2 \sum_{i=1}^{M} |Y_i(f)|^2}, \quad (1)$$

where $^*$ is the complex conjugate, $M$ is the number of windows, $X_i(f)$ and $Y_i(f)$ are the $i^{-\text{th}}$ window, Fourier Transforms of the signals. In cases where stimulation ($x[k]$) is deterministic, periodic and synchronized in each window ($X_i(f) = X(f)$, $\forall i$), the Eq. (1) can be simplified and called Magnitude-Squared Coherence [10]:

$$M\hat{S}C(f) = \frac{\left|\sum_{i=1}^{M} Y_i(f)\right|^2}{M\sum_{i=1}^{M} |Y_i(f)|^2}. \quad (2)$$

In case of absent response to stimulation, the numerator value tends to zero (for $M$ tending to infinity), thus, MSC tends to 0. On the other hand, in case of presence of an equal response in all windows, the value of MSC tends to 1.

For the Null Hypothesis (H0) of Absent Response, $M\hat{S}C(f)$ is distributed as a beta distribution, and the critical value for H0 is obtained with the following equation [19]:

$$MSC_{crit} = 1 - \alpha^{\frac{1}{M-1}}, \quad (3)$$

where $\alpha$ is the significance level. The signal is detected by rejecting H0, i.e., $MSC(f) > MSC_{crit}$.

### 2.2. Multiple magnitude-squared coherence

According to [16], the MMSC estimate between one periodic stimulus and the EEG of $N$ electrodes ($y_j[k]$, $j = 1,2,\ldots,N$) is given as

$$MM\hat{S}C(f) = \frac{V^H(f)\,S_{yy}^{-1}(f)\,V(f)}{M}, \quad (4)$$

where "$H$" denotes the Hermitian operator, $M$ is the number of data windows, $S_{yy}(f)$ is the estimate of the cross-spectral matrix of $N$ signals and

$$V(f) = \begin{bmatrix} \sum_{i=1}^{M} Y_{1i}^*(f) \\ \sum_{i=1}^{M} Y_{2i}^*(f) \\ \vdots \\ \sum_{i=1}^{M} Y_{Ni}^*(f) \end{bmatrix}, \quad (5)$$

where $Y_{ji}(f)$ – ($j = 1,2,\ldots$,N) – is the discrete Fourier Transform of the $i^{-th}$ signal segment in channel $j$.

Critical values with significance level α that constitute a threshold for detecting evoked responses may be obtained according to [19]:

$$MMSC_{crit} = \beta_{crit\alpha,N,M-N} \quad (6)$$

Where $\beta_{crit\alpha,\,N,\,M-N}$ is the inverse of the beta probability density function with shape parameters $a$ and $b$, evaluated for a significance level α. The detection of the presence of a response can be obtained by comparing the values of the $MMSC$ with the critical value ($MMSC(f_0) > MMSC_{crit}$).

## 3. Materials and methods

### 3.1. Experimental design

The experiments were performed in an acoustically isolated cabin in the Interdisciplinary Center for Signal Analysis (NIAS), Federal University of Viçosa. Total eight healthy subjects (2 female and 6 male) ages in between 20 and 43 years were included in the study. The subjects gave written consent to participate and the experiment was conducted according to a protocol approved by the local ethics committee (CEP/UFV No. 2.105.334). The volunteers were instructed to sit comfortably, keeping eyes closed and to not sleep. Each volunteer underwent 4 stimulation sessions (one for each carrier), each one lasting about 4.5 min.

The EEG was collected at electrodes FP1, FP2, Fz, F3, F4, F7, F8, Cz, C3, C4, T3, T4, Pz, P3, P4, T5, T6, O1 and O2 (10–20 International System) using the biological signal amplifier BrainNET BNT-36 (EMSA, Brazil) and Ag-AgCl electrodes. The reference electrode was Oz and the ground was the forehead (Fpz). The sampling frequency was set to 600 Hz, with the notch filter set to 60 Hz and band-pass filtering at 0.1–70 Hz.

A third-order Butterworth band-pass digital filter was offline applied with bandpass range from 30 to 50 Hz.

### 3.2. Auditory stimulation

The discrete-time expression of an AM tone is given as [20]:

$$x[n] = \frac{A \cdot sin[2\pi f_c n] \cdot (\lambda \cdot sin[2\pi f_m n] + 1)}{1 + \lambda} \quad (7)$$

where $A$ is the maximum amplitude, λ is the modulation depth, $f_c$ is the carrier frequency and $f_m$ is the modulating one. The auditory stimulation signals were generated with λ = 1. The $A$-values were adjusted to result in 50 dB SPL intensity. The calibration was performed using an artificial ear (BRÜEL & KJÄER – 4152 model) and a decibel meter (BRÜEL & KJÄER – 2260 model).

The stimulation protocol consisted of two distinct AM tones with same carrier frequency each one stimulating a different ear, with $f_m$ = 35 Hz for the left ear and with $f_m$ = 37 Hz for the right one. Such values were *a priori* corrected to result in an integer number of periods in each window in order to avoid spectral leakage in the detection of ASSRs. Considering a 1024-point window and the sampling frequency of the EEG acquisition system (600 Hz), the corrected modulating frequencies were 35.24 Hz (left ear) and 37.01 Hz (right ear). Four distinct carrier frequencies were used in the stimulation session: $f_c$ = 0.5, 1, 2 and 4 kHz. Such frequencies have been chosen to result in a separation of at least an octave among the carrier frequencies and to a gap between the modulating frequencies greater than 1.3 Hz. This care has been taken in order to avoid the ASSRs to interact with each other, as observed in [20].

The stimuli were presented throughout a shielded cable coupled to an insertion phone (Aearo Technologies – E-A-RTone 5A model).

### 3.3. Detection protocol

A sweep (M = 16 data windows of 1024 points each) was used to estimate the MSC and MMSC. If the detector value at a given frequency lied above its critical value for three consecutive sweeps, then detection in this frequency was achieved. Each sweep was averaged in time with the previous one before the application of the detectors. This procedure is similar to that in [21]. The significance level was 0.05.

### 3.4. Channels selection and performance measures

Since MSC is a univariate detector, the Cz channel was selected for MSC calculations [21]. In order to estimate MMSC, a set of signals must be provided. Thus, all possible combinations gathering from 2 to the 19 available electrodes were investigated in order to obtain the best electrode set, leading to 523,108 possible combinations.

Since the data from eight subjects were collected, and for each one four carriers were delivered binaurally, the number of delivered stimuli was NS = 64. Three performance measures were calculated for the detectors: detection ratio (DR), false alarm ratio (FAR) and mean detection time (DT).

The DR was the ratio between the number of detected responses (NDR) and the number of delivered stimuli. In order to calculate the FAR, eight distinct noise frequencies were chosen – 35.8315, 37.5938, 39.3560, 41.1182, 42.2930, 44.0552, 45.8174 and 47.5796 Hz – and the FAR was the ratio between the number of detected responses in the noise frequencies (DN) and the total number of noise frequencies (NN = 64). The mean DT was the average data length needed to detect the presence of all the possible the responses. In terms of equations:

$$DR = \frac{NDR}{NS} \quad (8)$$

$$FAR = \frac{DN}{NN} \quad (9)$$

$$DT = \frac{1}{NS}\sum_{1=1}^{NS} T, \quad (10)$$

where $T$ is the data length (in seconds) needed to detect the response according to the detection protocol.

The best subsets of EEG electrodes for MMSC were selected using a three-step procedure: first, it was chosen the subsets that obtained at least DR = 0.9. Secondly, only the subsets that obtained FAR less than 0.05 were maintained in the analysis. Finally, the DTs of the remaining subsets of electrodes were compared with the DTs obtained using MSC. The paired Wilcoxon test was applied to provide statistical method comparison.

**Table 1**
Performance measures for MMSC using "Fz and C4" and for MSC using "Cz". The two methods showed statistical difference according to the paired-Wilcoxon tests (p < 0.05).

| N | DR | FAR | DT (s) |
|---|---|---|---|
| Cz | 0.81 | 0.06 | 121.84 |
| Fz/C4 | 0.92 | 0.48 | 111.55 |

## 4. Results and discussion

A typical case of application of MSC (*N* = *1*) to the EEG data of a subject is shown in Fig. 1 below. In this figure, it can be seen that the modulations frequencies associated with stimuli are statistically different from noise, since their values are above the threshold.

In order to investigate the performance of the multivariate detector, we increased the number of signals used in the estimation of the detector (in this case, MMSC) and applied the detection protocol described in Section 3.3 for all available data and for all possible combinations of EEG electrodes. After counting the number of cases where the response was above the critical value, the number of electrode subsets that resulted in the same detection ratio was used to build the histogram displayed in Fig. 2.

It can be noted that the median of the DRs was around 0.7 and only a few cases surpass DR = 0.9.

In order to verify the detector's behavior as $N$ increases, the mean DR among the subjects was obtained and is shown in Fig. 3.

The results in Fig. 3 shows that the DR is expected to increase for $N$ < 6, i.e. the DR consistently drops for higher $N$. The results for $N$ > 16 are not shown just for visualization purposes.

The histogram in which more than 90% of the stimuli have been successfully detected are shown in Fig. 4. This figure was obtained by zooming Fig. 2 from 0.9 to 1 in the abscissa axis.

From the subsets described in Fig. 4, 74% obtained FAR ≤ 0.05. The best subset, i.e. the one with lower DT, is shown in Table 1. The performance of MSC using only Cz is also shown to allow comparison.

The multivariate approach increased DR in 13.58% while the DT was decreased in 8.45%. As expected, the false alarm ratio calculated over the noise frequencies were around 0.05. This result is in agreement with the detection theory of the constant false alarm ratio (CFAR) detectors [9], whose false alarm ratio must be equivalent to the significance level of the test.

## 5. Conclusion

The application of the MMSC in the ASSR evoked by AM tones detection proved to be more efficient when using more than one EEG signal. In fact, the increment in performance was 13.58% and 8.45% for DR and DT, respectively (an average improvement around 11%).

The best subset of EEG electrodes found was Fz and C4, which led to 0.92 detection ratio, with mean detection time of 111.55 s and

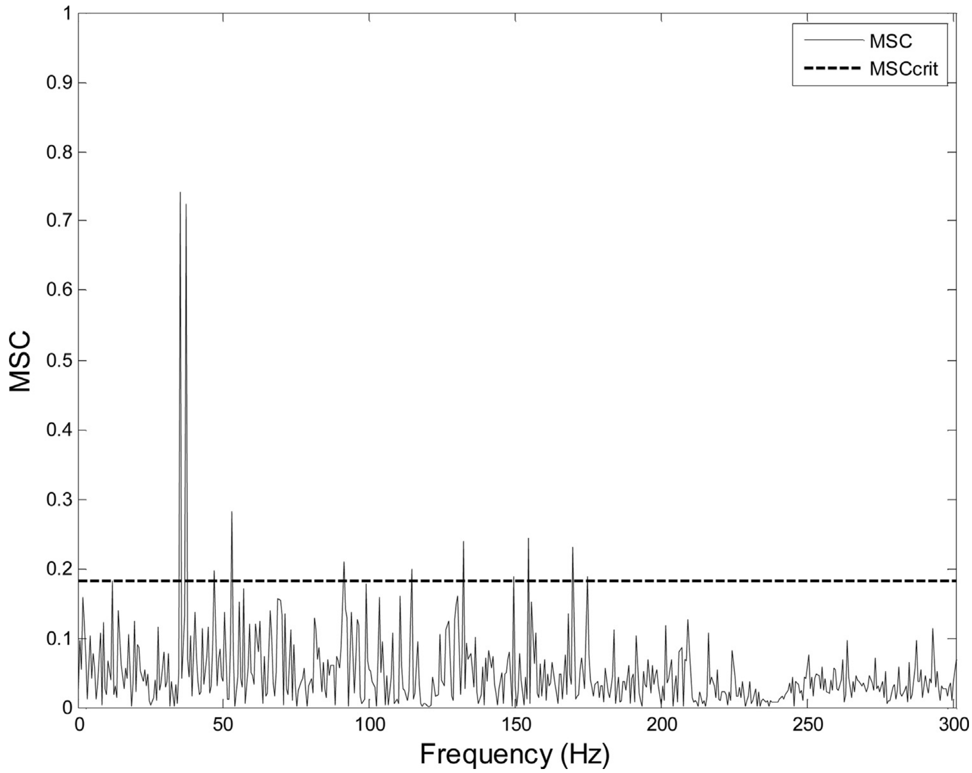


**Fig. 1.** Typical case of detection using MSC and Cz electrode. Continuos line is the MSC value as a function of frequency and dotted line is the critical value. The number of data windows was 36 and the significance level was 0.05.

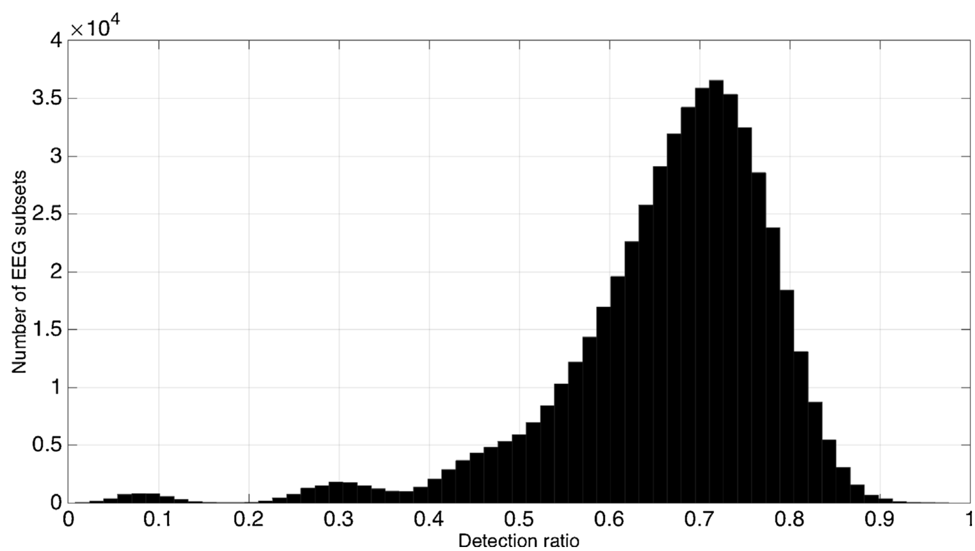


**Fig. 2.** Histogram showing the number of EEG subsets that obtained the same detection ratio.

0.48 of false alarm ratio. This performance was statistically significantly superior to that obtained by the univariate detector MSC. However, the multivariate approach must be compared with other univariate detectors, e.g. Component Synchrony Measure [17] or Mutual Information detector [22], which might require different data and statistical protocols.

It must be pointed out, however, that the number of signals for MMSC calculations should be higher than the number of data segments (otherwise a negative value for the shape parameter in the beta distribution appears, leading to meaningless MMSC values). This is an overall limitation of the technique, but it is negligible in the present work since the best results were found for $N = 2$.

Finally, the superior performance of this multivariate method in comparison with the univariate case may widen the applications of the coherence-based detectors to ASSRs. Besides the directly impact in auditory screenings, the technique proposed here could be useful in vision-free, brain-computer interfaces by allowing a higher number of commands to be executed, as well as in the evaluation of selective attention in the neurophysiological or clinical researches.

## Competing interests

No conflict of interest.

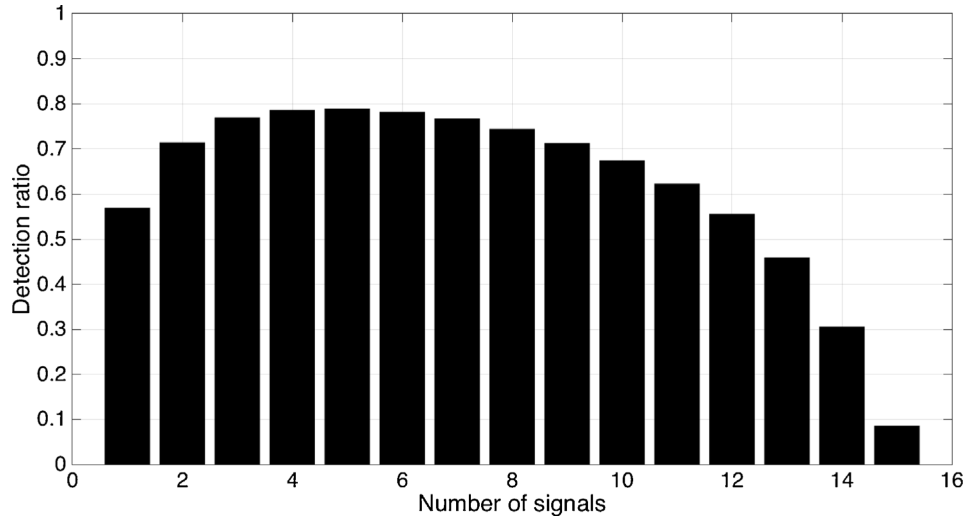


**Fig. 3.** Mean detection ratio as a function of number of signals.

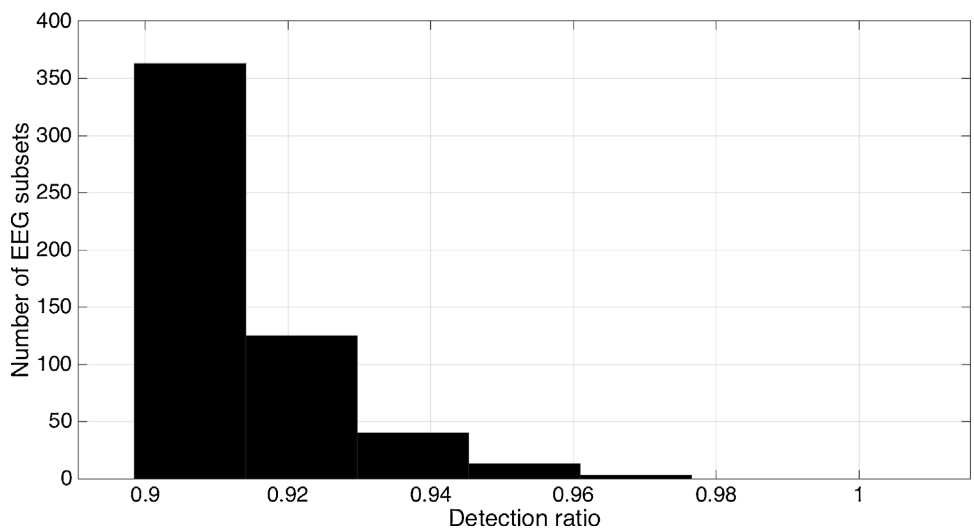


**Fig. 4.** Histogram showing the number of electrodes subsets that obtained DR > 0.9.

## Ethical approval

Work approved by the Local Ethics Committee (CEP/UFV No. 2.105.334).

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